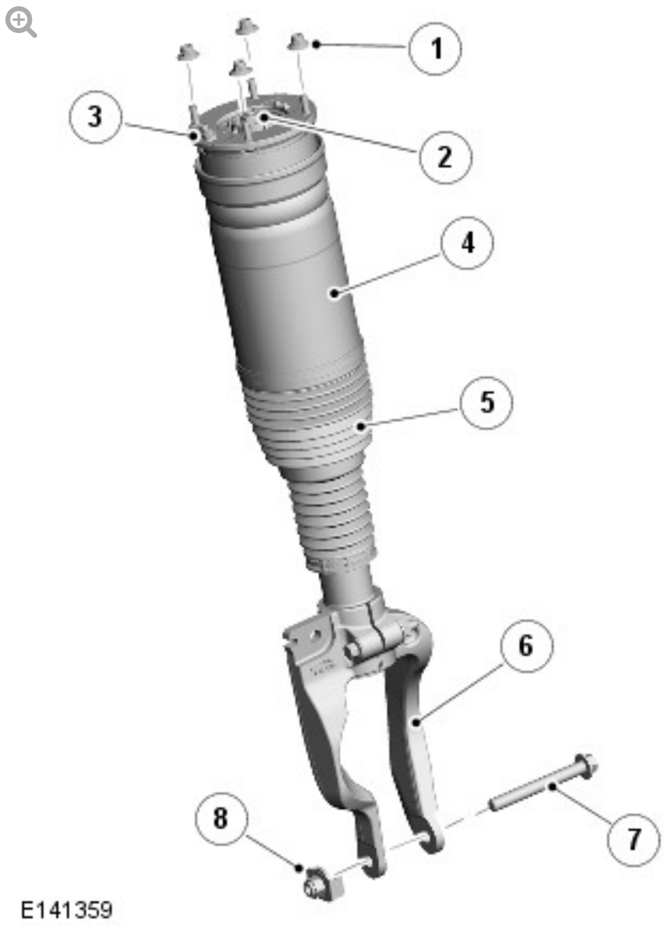
FRONT SUSPENSION

DESCRIPTION AND OPERATION

DESCRIPTION

AIR SPRING AND DAMPER MODULE



ITEM	DESCRIPTION
1	Nut (damper to vehicle body)

2	Nut (damper rod)
3	Air connector
4	Air spring and damper assembly
5	Gaiter
6	Damper yoke
7	Bolt (damper yoke to rear lower arm)
8	Nut (damper yoke to rear lower arm)

DAMPER

The damper assembly is a mono tube design with an air spring. The lower end of the damper is fitted with a yoke which is attached to the rear lower arm. The damper yoke also includes a location hole for the stabilizer link.

The damper functions by restricting the flow of hydraulic fluid through internal galleries within the damper. The damper rod moves axially within the damper, its movement limited by the flow of fluid through the galleries, providing damping of undulations in the terrain. The damper rod is sealed at its exit point from the damper body to maintain the fluid within the unit and to prevent the ingress of dirt and moisture. The seal also incorporates a wiper to keep the rod clean.

AIR SPRING

The air spring comprises an aluminium restraining cylinder, top mount, spring aid, air sleeve, an inner support sleeve aluminium piston and a lower isolator. Elastomeric gaiters are fitted top and bottom. The upper gaiter is a self-supporting cover. The lower gaiter is of a convoluted design and fitted to the restraining cylinder by a metal clip and is attached to the damper by a plastic ring clipped to the damper.

The air sleeve is made from a flexible rubber material which allows the sleeve to roll up and down the air spring piston as the vehicle changes height. The air sleeve is attached to the air spring piston and top mount by crimp rings which provide an air tight seal, surrounded by the restraining cylinder. The air spring piston mounts to a spring isolator which contains an

O-ring sealing the piston assembly to the damper body. The piston assembly is positively attached to the seat with retaining spigots. The top of the air sleeve is crimped to the top mount which attaches to the body with four integral studs and self-locking nuts.

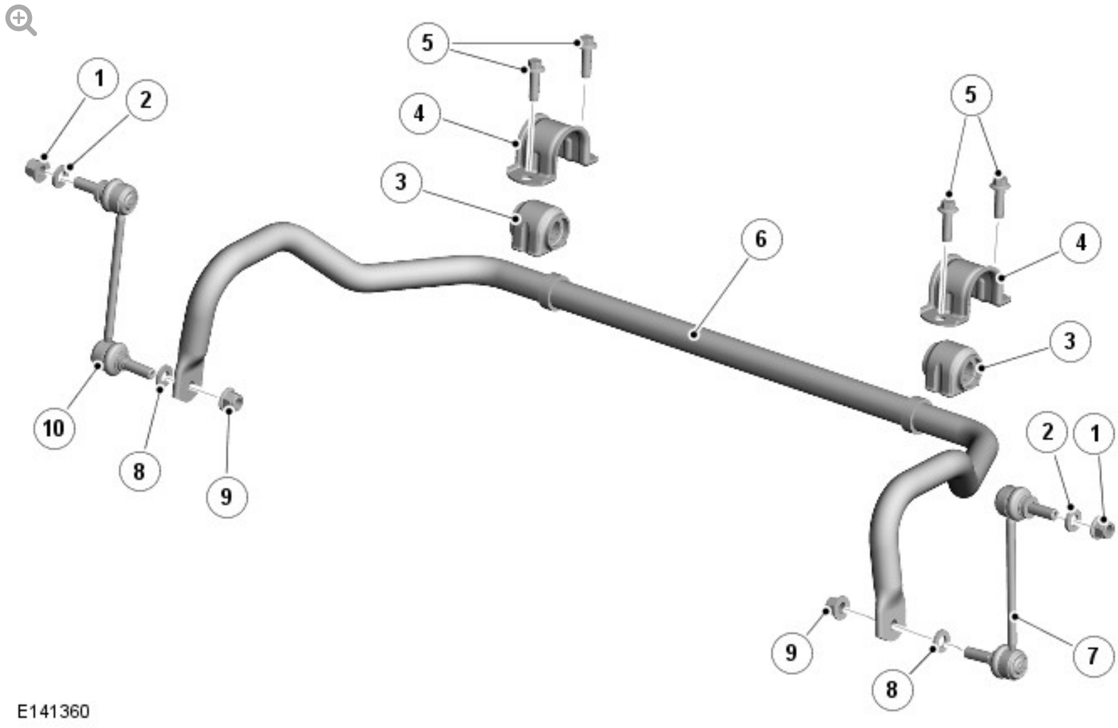
A spring aid is fitted to the top mount concentric with the damper rod and prevents the top mount contacting the bump cap fitted to the top of the damper during full suspension compression. The lower end of the air spring is located over the damper body and seats on a fabricated seat on the damper body. The damper rod is located through a central hole in the top mount. The rod is threaded at its outer end. A nut secures the air spring to the damper rod.

The top mount is an integral part of the air spring and is fitted with a bush and a dynamic seal. The top mount is secured to the damper rod with a nut. The top mount attaches to a suspension tower on the body with four integral studs and self-locking nuts. The top mount also incorporates an air fitting which allows for the attachment of the air pipe.

NOTE:

> Air spring internal components are not serviceable.

STABILIZER BAR AND LINKS



ITEM	DESCRIPTION
1	Nut (stabilizer link to damper yoke)
2	Washer (stabilizer link to damper yoke)
3	Bush (stabilizer bar to subframe)
4	Bracket (stabilizer bar to subframe)
5	Bolt (stabilizer bar bracket to subframe)
6	Stabilizer bar
7	Stabilizer link (left)
8	Washer (stabilizer link to stabilizer bar)
9	Nut (stabilizer link to stabilizer bar)
10	Stabilizer link (right)

Vehicles with the Dynamic Response system use an active stabilizer bar
For additional information, refer to: Ride and Handling Optimization (204-06 Ride and Handling Optimization, Description and Operation).

The stabilizer bar is fabricated from induction hardened, solid spring steel bar. The stabilizer bar operates, via a pair of links, from their attachment to the lower arm.

The stabilizer bar is mounted on the subframe’s rear cross member and is attached with two, Teflon lined bushes. Brackets, which are pressed onto the bushes, are attached to the cross member with bolts. The stabilizer bar has crimped, 'anti-shuffle' collars pressed in position on the inside edges of the bushes. The collars prevent sideways movement of the stabilizer bar.

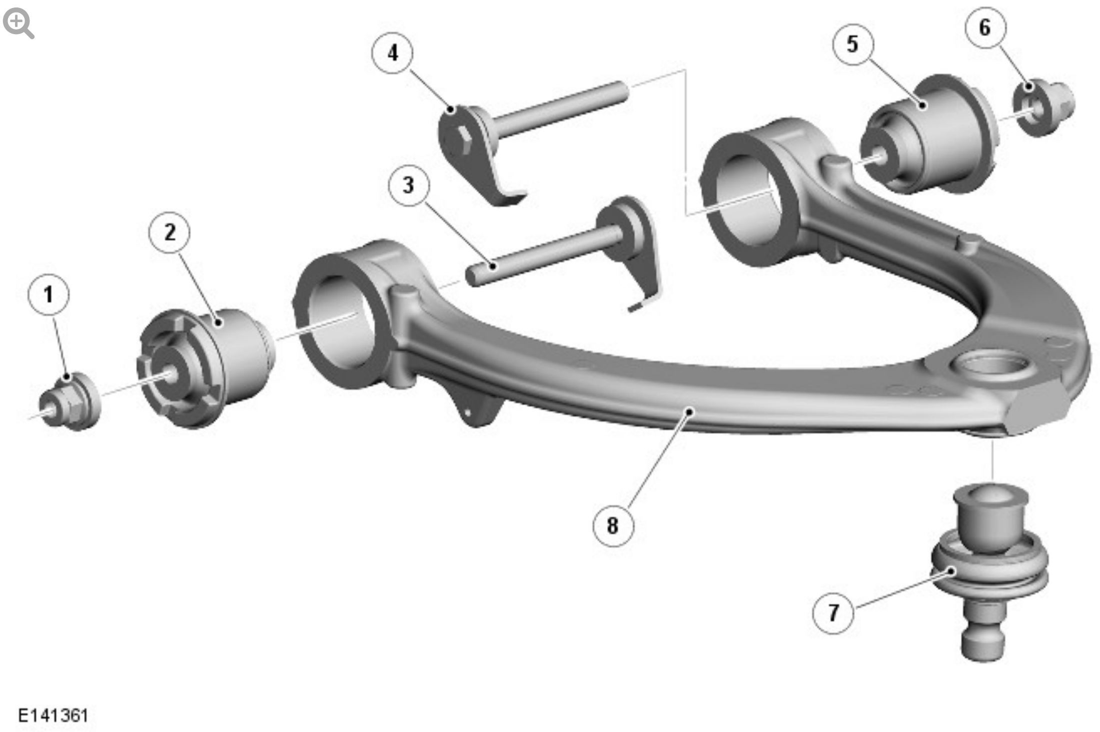
The links have a ball joint at each end, are fitted with hardened steel washers at both ends. The top ball joint is attached to the link at 90 degrees to the link axis. The ball joint is located in a hole in the damper yoke and secured with the hardened steel washer on one side of the interface and a self-locking nut on the other. The bottom ball joint is attached to the link at 90 degrees to the link axis. The ball joint is located in a hole in the end of the stabilizer bar and secured with the hardened steel washer on one side of the interface and the self-locking nut on other

It is important that the hardened steel washer is in the correct position between the stabilizer bar and the link ball joint and the damper yoke and the link ball joint and the correct, hardened washer is fitted.

CAUTION:

Failure to fit the washer or using an incorrect washer will result in relaxation of the torque on the self-locking nut and damage will be caused to the stabilizer bar, link and damper yoke.

UPPER ARM



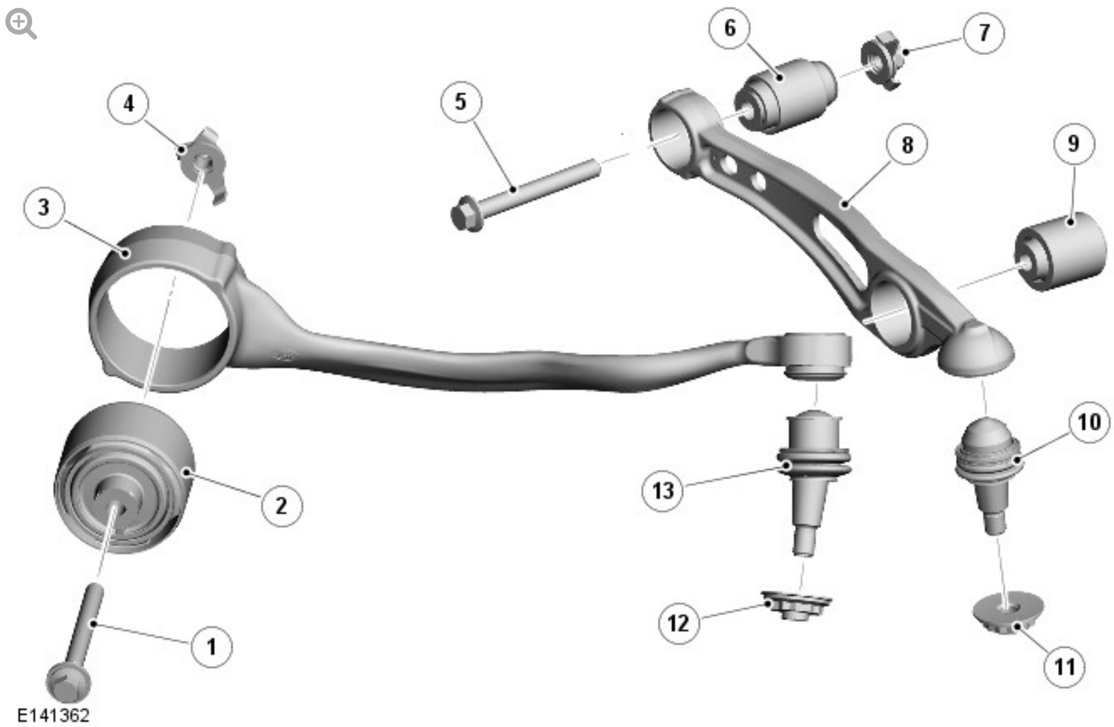
ITEM	DESCRIPTION
1	Nut (front bush)
2	Bush - front
3	Bolt (front bush)
4	Bolt (rear bush)
5	Bush - rear
6	Nut (rear bush)
7	Ball joint (upper arm to wheel knuckle)
8	Upper arm

The upper arm assembly comprises: the upper arm, two bushes and a ball joint. The upper arm is made from forged and machined steel. Its outer end has a hole to accept the ball joint.

The inner end of the arm has two bush housings which are machined in the arm. A bush is pressed into each of the housings. The bushes are located in the body structure and are secured with bolts and self-locking nuts through metal inserts in the centre of the bushes.

The ball joint is pressed into the upper arm and is secured with an interference fit. The top face of the ball joint has two semi-circular cut-outs. One of these cut-outs must be aligned with the small indentation in the upper arm to ensure the correct operation of the ball joint.

LOWER ARMS



ITEM	DESCRIPTION
1	Bolt (front lower arm to subframe)
2	Bush (front lower arm to subframe)
3	Front lower arm
4	Nut (front lower arm to subframe)
5	Bolt (rear lower arm to subframe)
6	Bush (rear lower arm to subframe)
7	Nut (rear lower arm to subframe)
8	Rear lower arm
9	Bush (rear lower arm to damper yoke)
10	Ball joint (rear lower arm to wheel knuckle)

11	Nut (rear lower arm to wheel knuckle)
12	Nut (front lower arm to wheel knuckle)
13	Ball joint (front lower arm to wheel knuckle)

The lower assembly comprises: two independent arms, with each arm, consisting of a bush and ball joint.

Both arms are made from forged and machined steel. An aperture in the rear arm provides for the attachment of the damper yoke bush.

Each lower arm is fitted with a bush which is secured with a bolt to the subframe, with bolts and self-locking nuts through metal inserts in the centre of the bushes. The outer-end of the arms are fitted with a tapered ball joint which attach to the wheel knuckle.

The ball joint is pressed into the upper arm and is secured with an interference fit. The top face of the ball joint has two semi-circular cut-outs. One of these cut-outs must be aligned with the small indentation in the upper arm to ensure the correct operation of the ball joint.

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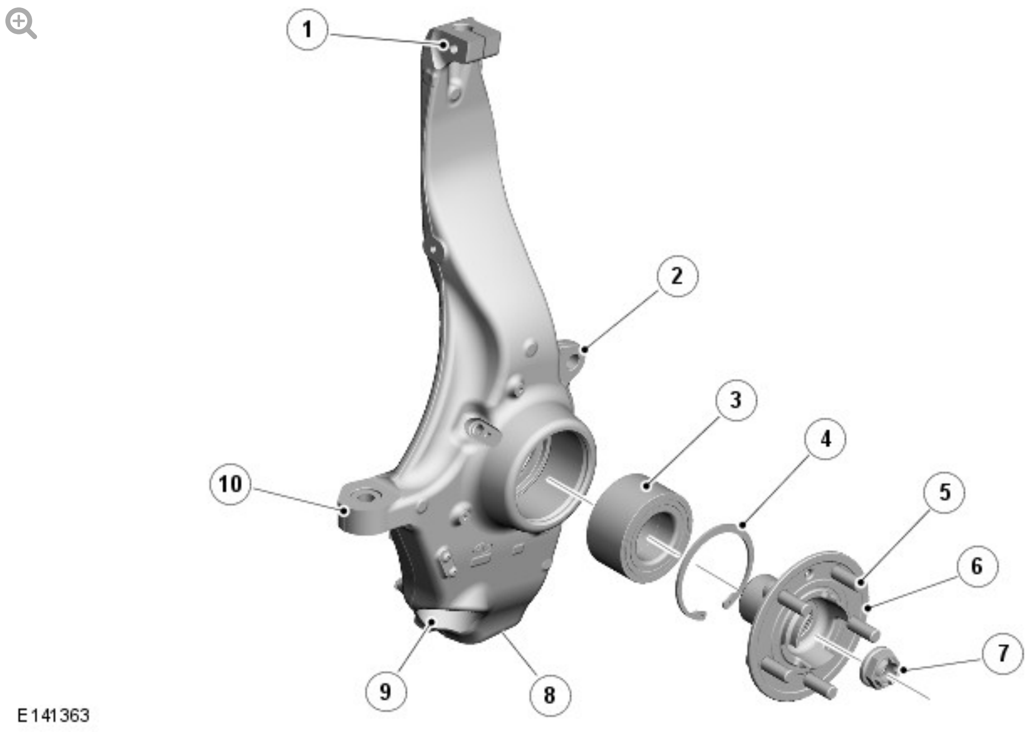
WHEEL KNUCKLE, HUB AND BEARING ASSEMBLY



ITEM	DESCRIPTION
1	Upper arm attachment location
2	Brake caliper attachment location
3	Bearing (wheel hub)
4	Circlip (wheel hub bearing)
5	Wheel stud
6	Wheel hub
7	Nut (wheel hub)
8	Rear lower arm attachment location
9	Front lower arm attachment location
10	Track rod end attachment location

The wheel knuckle is a machined aluminum forging which is located between the ball joints of the upper and lower arms. A cast boss on the front edge of the knuckle provides for the attachment of the steering gear, tie rod ball joint.

The wheel hub comprises: the wheel hub housing, taper roller bearing. Five studs are pressed into the wheel hub and provide for the attachment of the road wheel.

The wheel hub housing is a machined forging which houses the taper roller bearing.

The wheel hub has a splined centre bore which mates with corresponding splines on the half-shaft. Rotation of the half-shaft is passed, via the splines, to the wheel hub which rotates on the taper roller bearing.